

Defining Problem Statements Summary

Smart-Lender Project Documentation Suite

Project Name	Smart Lender - AI-Powered Loan Approval Prediction System
Document Type	Defining Problem Statements Summary
Project Phase	Phase 1: Brainstorming & Ideation
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Organization / College	Vishnu Institute of Technology
Status / Version	Approved / v1.0 (Exact Replica Format)

Step 1: Customer Problem Profile Framework

Persona Grid Option	Detailed Student Team Observations (Smart Lender Project)
I am...	A credit appraisal manager handling a high quantity of retail loan files daily.
I'm trying to...	Evaluate customer applications based on historical logs and financial matrices.
But...	The auditing steps are slow, error-prone, and introduce human inconsistencies.
Because...	There is no automated tool to integrate categorical encoders, SMOTE scaling, and unified classifiers in a single dashboard.
I feel...	Fatigued, concerned about credit default parameters, and worried about missing bank SLAs.

Step 2: Final Problem Statement Matrix

Core Problem Domain	Target Users Impacted	Root Bottleneck Cause	Target Metric of Success
Retail Lending Compliance	Bank credit analysts, underwriters, and retail loan applicants	Manual evaluations of variables like debt levels and credit histories	Decrease appraisal decisions time to under 2 seconds while maintaining accuracy > 84%